THE UNIVERSITY OF HONG KONG
FACULTY OF BUSINESS AND ECONOMICS

MSBA7021 Optimization and Reinforcement Learning

GENERAL INFORMATION

Instructor: Prof. Zhang, Wei
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Consultation times: By appointment

Tutor: Miss HE, Guoxing

Pre-requisites: None (Prior knowledge required: calculus & linear algebra; probability & statistics; Python coding)
Co-requisites: None
Mutually exclusive: None

Reference Book:
There is no required textbook for this course. Required readings will be posted on the website.
Optional reference book: Reinforcement Learning: An Introduction. Second Edition, by Sutton and Barto. The MIT Press.
The following programming languages are used in this course: Python.

COURSE DESCRIPTION

This course introduces students to the fundamental theories and computational methods underlying optimization and reinforcement learning—two core frameworks in advanced analytics and artificial intelligence. It focuses on understanding how intelligent systems learn to make sequential decisions through interactions with their environment, enabling adaptive and autonomous behavior across dynamic business contexts. The course begins with foundational topics and builds towards more advanced concepts and techniques that form the backbone of modern machine learning algorithms applied in domains such as recommendation systems, operations optimization, and strategic planning. In addition, the course bridges theory and practice by introducing insights from psychology and neuroscience that inspire learning mechanisms in artificial agents. Through hands-on exercises, coding demonstrations, and a final group project, students gain practical experience implementing and applying these algorithms using Python. By the end of the course, participants will have developed a solid conceptual and technical foundation in optimization and reinforcement learning for data-driven intelligence and adaptive system design.

COURSE OBJECTIVES

1. To provide students a broad introduction to optimization methodologies in business analytics
2. To enable students to understand state-of-the-art optimization methods for decision making under uncertainty
3. To provide students comprehensive toolsets to transform valuable data to valuable decisions
4. To allow students to solve decision making problems under uncertainty in a data-rich business environment using software packages

PROGRAMME LEARNING OUTCOMES

PLO1: Acquisition and internalization of knowledge of the programme discipline

PLO2: Application and integration of knowledge

PLO3: Inculcating professionalism and leadership

PLO4: Developing global outlook

PLO5: Mastering communication skills

COURSE LEARNING OUTCOMES

Course Learning Outcomes	Aligned Program Learning Outcomes
CLO1: Identify a business decision making problem and formulate it into an optimization problem	PLO 1
CLO2: Select and use effective methods to address the major challenges	PLO 2
CLO3: Use analytical tools and software packages to solve the decision-making problem	PLO 2, 4
CLO4: Communicate the solution effectively	PLO 3, 5

COURSE TEACHING AND LEARNING ACTIVITIES			
Course Teaching and Learning Activities		**Expected contact hour**	**Study Load (% of study)**
T&L1. Interactive lectures		30	25%
T&L2. Assignments & Tutorials		60	50%
T&L3. Self-study, group discussion		30	25%
	Total	120	100%
Assessment Methods	**Brief Description (Optional)**	**Weight**	**Aligned Course Learning Outcomes**
A1. In-class and forum participation	Discussions	20%	CLO 1, 2, 4
A2. Assignments	Effort and accuracy	60%	CLO 1, 2, 3
A3. Group Project	Effort and accuracy	20%	CLO 1, 2, 3, 4
	Total	100%	

STANDARDS FOR ASSESSMENT	
Course Grade Descriptors	
A+, A, A-	• Demonstrate a strong understanding of all relevant knowledge • Handling questions professionally • High participation in discussions • Present arguments that have an element of originality • Achieve a standard of excellent performance in the exams with very accurate computation and very good analytical and problem-solving skills • Excellent performance in assignments
B+, B, B-	• Demonstrate a good understanding of all relevant knowledge • Handling questions in a logical way • Good participation in discussions • Present arguments that go beyond the lecture and textbook • Achieve a standard of good performance in the exams with accurate computation and good analytical and problem-solving skills • Good performance in assignments
C+, C, C-	• Demonstrate a basic understanding of the concepts involved • Fairly address questions as set • Some participation in discussions • Present arguments in a well-structure manner • Meet a standard of acceptable performance in the exams with reasonably accurate computation and acceptable analytical and problem-solving skills • Acceptable performance in assignments
D+, D	• Demonstrate a minimum understanding of the concepts involved • Barely address questions as set • Minimal or no participation in discussions • Present arguments in a marginally acceptable manner • Meet a standard of marginally acceptable performance in the exams with some errors in computation and barely adequate analytical and problem-solving skills • Marginally acceptable performance in assignments
F	• Demonstrate a poor understanding of the concepts involved • Unable or unwilling to handle questions • Minimal or no participation in discussions • Present arguments poorly • Fail to meet a standard of passing the exams with major errors in computation and inadequate analytical and problem-solving skills • Poor performance in assignments
Assessment Rubrics for Participation	
A+, A, A-	• High participation in discussions • Always attend in-class discussions • Demonstrate a strong understanding of all relevant knowledge • Handling questions professionally • Present arguments that have an element of originality

	• Respect others and follow the class rules (no chatting and do not use cell phone)
B+, B, B-	• Good participation in discussions • Often attend in-class discussions • Demonstrate a good understanding of all relevant knowledge • Handling questions in a logical way • Present arguments that go beyond the lecture and textbook • Respect others and follow the class rules (no chatting and do not use cell phone)
C+, C, C-	• Some participation in discussions • Sometimes attend in-class discussions • Demonstrate a basic understanding of the concepts involved • Fairly address questions as set • Present arguments in a well-structure manner • Respect others and follow the class rules (no chatting and do not use cell phone)
D+, D	• Minimal or no participation in discussions • Rarely attend in-class discussions • Demonstrate a minimum understanding of the concepts involved • Barely address questions as set • Present arguments in a marginally acceptable manner • Respect others and follow the class rules (no chatting and do not use cell phone)
F	• Minimal or no participation in discussions • Almost never attend in-class discussions • Demonstrate a poor understanding of the concepts involved • Unable or unwilling to handle questions • Present arguments poorly • Behave poorly in class (often chatting with others, using cell phones, or being late)
Assessment Rubrics for Assignments	
A+, A, A-	• Demonstrate a strong understanding of all relevant knowledge • Present arguments that have an element of originality • Achieve a standard of excellent performance in the assessments with very accurate computation and very good analytical and problem solving skills
B+, B, B-	• Demonstrate a good understanding of all relevant knowledge • Present arguments that go beyond the lecture and textbook • Achieve a standard of good performance in the assessments with accurate computation and good analytical and problem solving skills
C+, C, C-	• Demonstrate a basic understanding of the concepts involved • Present arguments in a well-structure manner • Meet a standard of acceptable performance in the assessments with reasonably accurate computation and acceptable analytical and problem solving skills
D+, D	• Demonstrate a minimum understanding of the concepts involved • Present arguments in a marginally acceptable manner • Meet a standard of marginally acceptable performance in the assessments with some errors in computation and barely adequate analytical and problem solving skills
F	• Demonstrate a poor understanding of the concepts involved • Present arguments poorly • Fail to meet a standard of passing the assessments with major errors in computation and inadequate analytical and problem solving skills
Assessment Rubrics for Project Report	
A+, A, A-	Provides persuasive original, insightful and well-reasoned recommendations that clearly follow from the analysis and effectively address the key issue
B+, B, B-	Provides well-reasoned recommendations that follow from the analysis and address the key issue
C+, C, C-	Conducts adequate analysis and provides reasoned recommendations that address the key issue
D+, D	Offers weak recommendations, which do not address key issue
F	Fails to identify key issue and/or provide adequate analysis and/or recommendations

COURSE CONTENT AND TENTATIVE TEACHING SCHEDULE

- Session 01 (Jan 23): Introduction to Reinforcement Learning
- Session 02 (Jan 27): Multi-Armed Bandits
- Session 03 (Jan 30): Dynamic Programming
- Session 04 (Feb 03): Markov Analysis
- Session 05 (Feb 13): Markov Decision Process
- Session 06 (Feb 17): Model-Based Algorithms
- Session 07 (Feb 20): Model-Free Learning
- Session 08 (Feb 24): Monte Carlo Methods
- Session 09 (Feb 27): Approximation Methods
- Session 10 (Mar 03): Psychology and Neuroscience
- Session 11 (Mar 06): Project Presentation

REQUIRED/RECOMMENDED READINGS & ONLINE MATERIALS (e.g. journals, textbooks, website addresses etc.)

To be announced on Moodle.

MEANS/PROCESSES FOR STUDENT FEEDBACK ON COURSE

Conducting mid-term survey in additional to SETL around the end of the semester

COURSE POLICY (e.g. plagiarism, academic honesty, attendance, etc.)

An orderly learning environment is extremely important. Disruptive behaviors, such as chatting during lectures, arriving to classes late, and abuses of mobile devices, are unacceptable. Students who are responsible for any of these actions will be subject to academic penalty and will be asked to leave the classroom.

Academic dishonesty includes cheating, plagiarism, unauthorized collaboration, falsifying academic records, and any act designed to avoid participating honestly in the learning process. Any such dishonesty will result in an F grade.

ADDITIONAL COURSE INFORMATION (e.g. e-learning platforms & materials, penalty for late assignments, etc.)

No late assignment will be accepted.

Lecture notes and self-learning materials will be uploaded on Moodle.

The instructor reserves all the rights to make necessary changes to the syllabus. These changes will be announced as soon as possible.